

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semester Toets 2

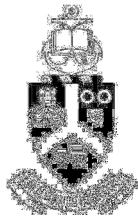
Kopiereg voorbehou

Semester Test 2

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Module EBN111
Elektrisiteit en Elektronika
14 Mei 2014

Module EBN111
Electricity and Electronics
14 May 2014



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: *Exam information:*

Maksimum punte: <i>Maximum marks:</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute <i>90 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		19 (Antwoordblad ingesluit / <i>Including answer sheet</i>)	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.
- Tensy anders aangedui, aanvaar dat alle operasionele versterkers ideaal is.**
Unless otherwise specified, assume all operational amplifiers are ideal.
- Polêre vorm: $A\angle\theta$; Reghoekige vorm: $a + jb$**
Polar form: $A\angle\theta$; Rectangular form: $a + jb$

Dosent(e): <i>Lecturer(s):</i>	1 JD le Roux
	2 DV Bhatt

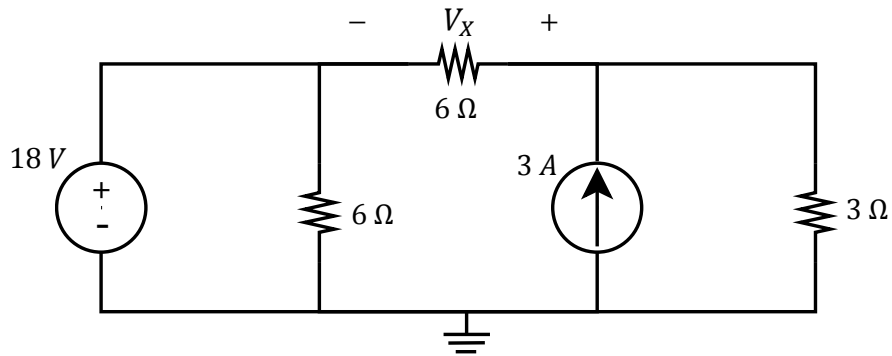
VRAAG/QUESTION 1 [25]

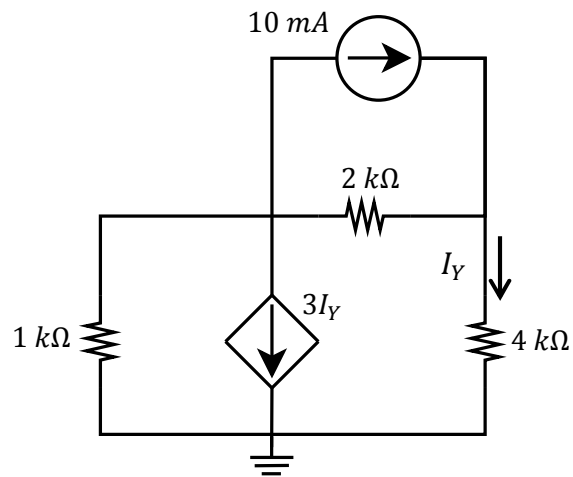
Vraag 1.1 [2]

Maak gebruik van superposisie en bepaal die bydrae tot V_X slegs as gevolg van die 3 A stroombron.

Question 1.1 [2]

Use superposition and determine the contribution to V_X by the 3 A current source only.



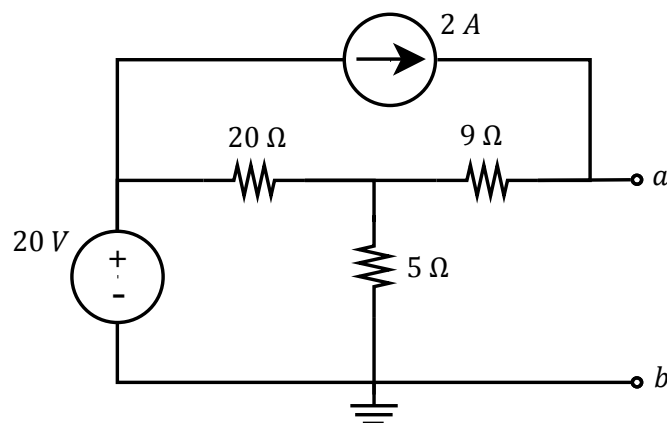
Vraag 1.2 [2]Maak gebruik van brontransformatie en bepaal I_Y .**Question 1.2 [2]**Make use of source transformation and determine I_Y .

Vraag 1.3 [4]

Bepaal die Thevenin ekwivalente spanning V_{TH} en weerstand R_{TH} tov terminale a - b .

Question 1.3 [4]

Determine the Thevenin equivalent voltage V_{TH} and resistance R_{TH} wrt terminals a - b .

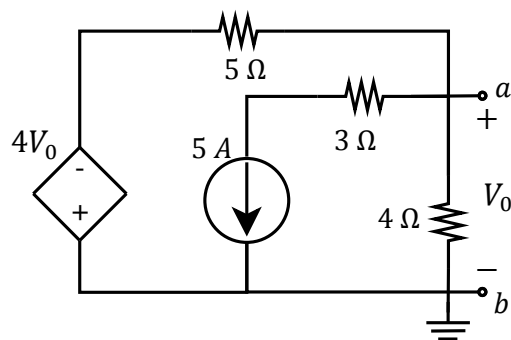


Vraag 1.4 [4]

Bepaal die Norton ekwivalente stroom I_N en weerstand R_N tov terminale $a-b$.

Question 1.4 [4]

Determine the Norton equivalent current I_N and resistance R_N wrt terminals $a-b$.



Vraag 1.5 [3]

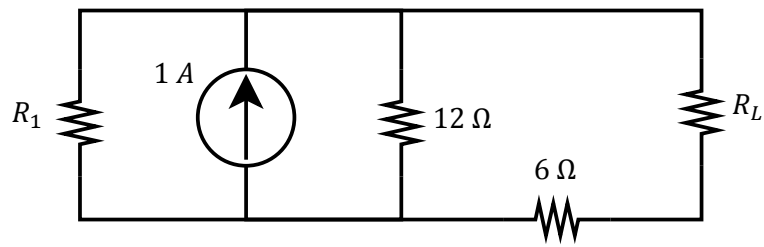
Vir maksimumdrywing oordrag aan R_L , bepaal

- a) R_L , indien $R_1 = 6\ \Omega$ [1],
b) P_{max} , indien $R_1 = \infty\ \Omega$ [2].

Question 1.5 [3]

For maximum power transfer to R_L , determine:

- a) R_L , if $R_1 = 6\ \Omega$ [1],
b) P_{max} , if $R_1 = \infty\ \Omega$ [2].

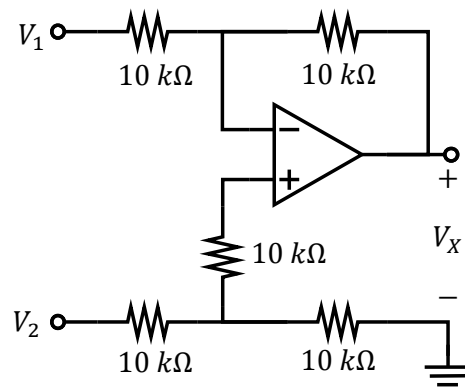


Vraag 1.6 [3]

- a) Indien $V_1 = 10\text{ V}$ en $V_2 = 0\text{ V}$, bepaal V_X .
b) Indien $V_1 = 0\text{ V}$ en $V_2 = 10\text{ V}$, bepaal V_X .
c) Indien $V_1 = 10\text{ V}$ en $V_2 = 10\text{ V}$, bepaal V_X .

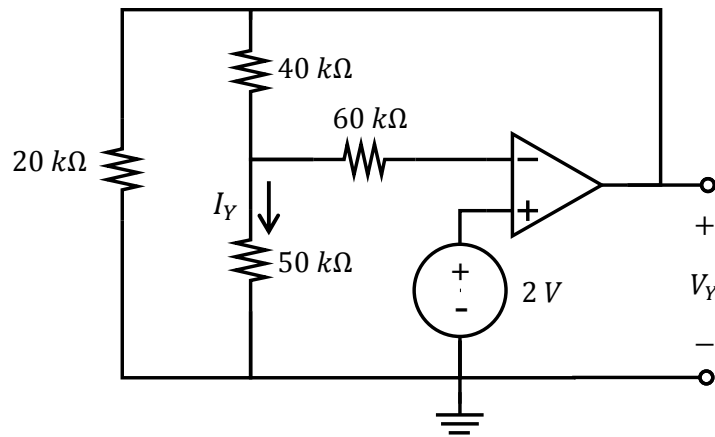
Question 1.6 [3]

- a) If $V_1 = 10\text{ V}$ and $V_2 = 0\text{ V}$, determine V_X .
b) If $V_1 = 0\text{ V}$ and $V_2 = 10\text{ V}$, determine V_X .
c) If $V_1 = 10\text{ V}$ and $V_2 = 10\text{ V}$, determine V_X .



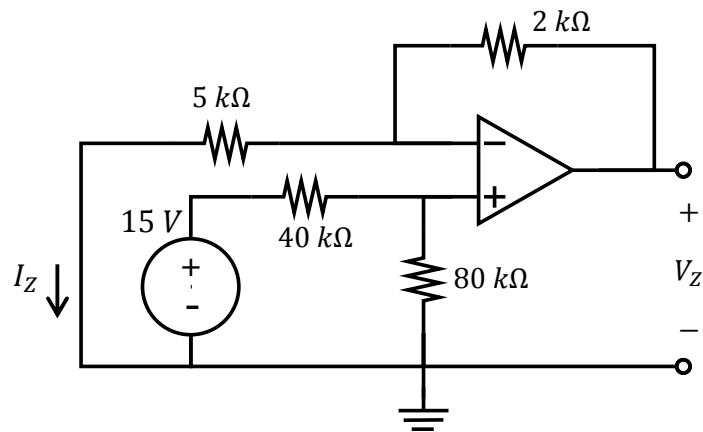
Vraag 1.7 [2]
Bepaal V_Y en I_Y .

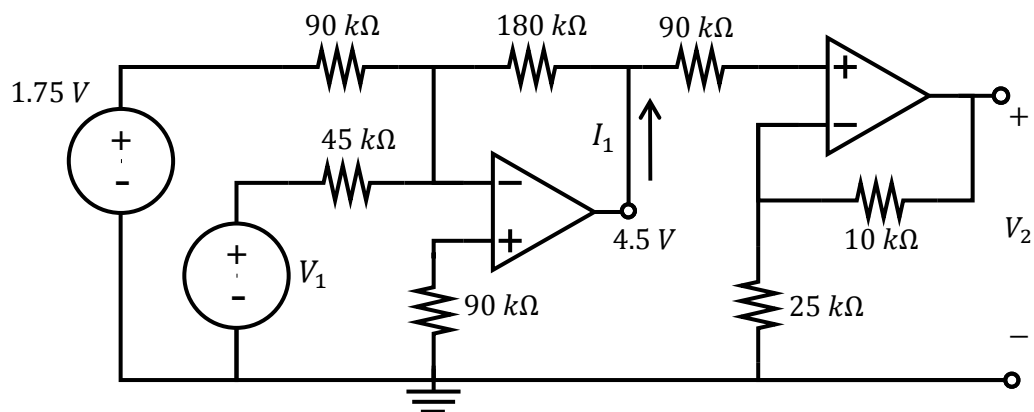
Question 1.7 [2]
Determine V_Y and I_Y .



Vraag 1.8 [2]
Bepaal V_Z en I_Z .

Question 1.8 [2]
Determine V_Z and I_Z .



Vraag 1.9 [3]Bepaal V_1 , V_2 en I_1 .**Question 1.9 [3]**Determine V_1 , V_2 and I_1 .

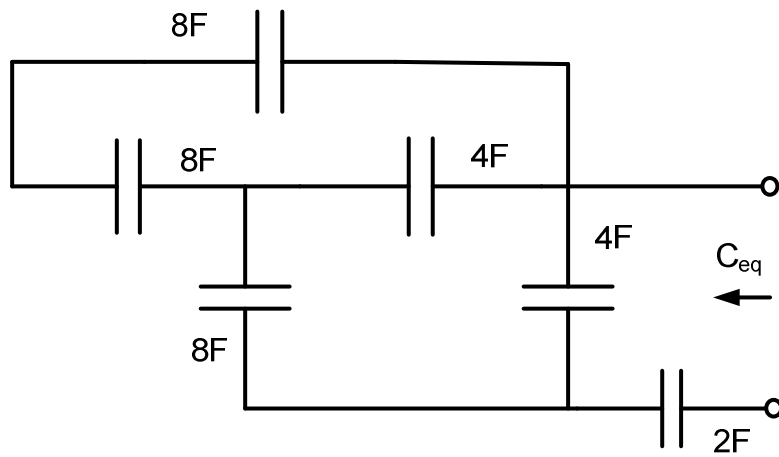
VRAAG/QUESTION 2 [20]

Vraag 2.1 [2]

Bepaal die ekwivalente kapasitansie C_{eq} .

Question 2.1 [2]

Calculate the equivalent capacitance C_{eq} .



Vraag 2.2 [2]

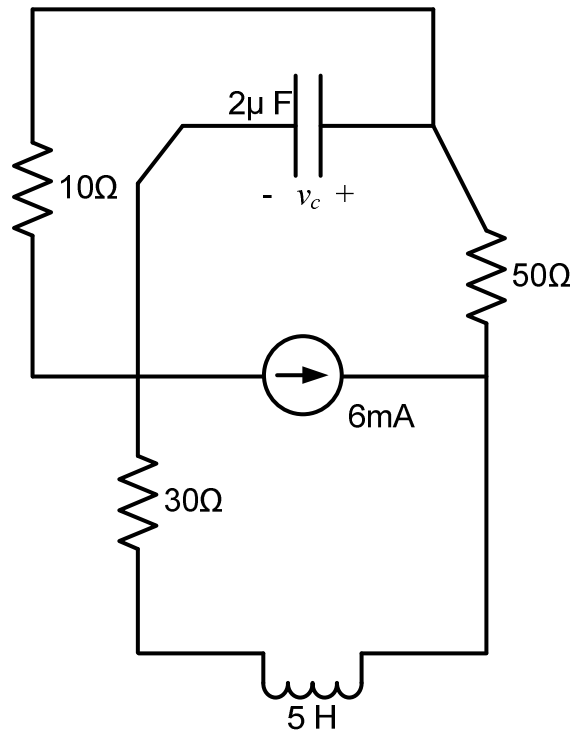
Bepaal:

- a) die spanning v_C oor die $2\mu\text{F}$ kapasitor, en
- b) die energie w_L gestoor in die 5H induktor.

Question 2.2 [2]

Determine:

- a) the voltage v_C across the $2\mu\text{F}$ capacitor, and
- b) the energy w_L stored in the 5H inductor.



Vraag 2.3 [2]

Twee induktore van 5 H en 10 H word in serie met 'n spanningsbron $v_s(t) = 2t$ V gekoppel.

(Aanvaar $i(0) = 0$ A.)

- a) Bereken die stroom $i(t)$ in die kringbaan. Gee jou finale antwoord in ***mA***.
- b) Bereken die energie gestoor in die 10 H inductor by $t = 1$ s. Gee jou finale antwoord in ***mJ***.

Question 2.3 [2]

Two inductors of 5 H and 10 H are connected in series with a voltage source $v_s(t) = 2t$ V.

(Assume $i(0) = 0$ A.)

- a) Calculate the current $i(t)$ flowing through circuit. Give your final answer in ***mA***.
- b) Calculate the energy stored in the 10 H inductor at $t = 1$ s. Give your final answer in ***mJ***.

Vraag 2.4 [6] a) Bereken die frekwensie van die sinusied in Hz: $i(t) = -15 \sin(502.7t - 20^\circ)$ A b) Bepaal die fasehoek ϕ nadat die sinusied $i(t) = 12 \sin(\omega t - 70^\circ)$ omgeskakel is na $i(t) = A \cos(\omega t + \phi)$ waar $A > 0$ en $\phi \leq 180^\circ$. c) Gee die fasor $\bar{V}$ vir die sinusied $v(t) = -5 \cos(\omega t + 125^\circ)$ V. (Onthou $r > 0$ en $\theta \leq 180^\circ$ vir $r \angle \theta$.) d) Watter van die volgende twee sinusiede is voorlopend en met hoeveel grade? $v_1(t) = -2.5 \cos(\omega t + 70^\circ)$ V $v_2(t) = 3 \sin(\omega t - 70^\circ)$ V e) Bepaal die sinusied $i(t)$ van die stroomfasor $\bar{I}$, indien $\bar{I} = j3e^{j15^\circ}$ A en $\omega = 50$ rad/s. Gee die finale antwoord in cosinus vorm met 'n positiewe amplitude en 'n fasehoek tussen $\pm 180^\circ$.	Question 2.4 [6] a) Calculate the frequency of the sinusoid in Hz: $i(t) = -15 \sin(502.7t - 20^\circ)$ A b) What is the phase angle ϕ after converting the sinusoid $i(t) = 12 \sin(\omega t - 70^\circ)$ to $i(t) = A \cos(\omega t + \phi)$ where $A > 0$ and $\phi \leq 180^\circ$? c) Write the phasor representation $\bar{V}$ of the sinusoid $v(t) = -5 \cos(\omega t + 125^\circ)$ V. (Remember, $r > 0$ and $\theta \leq 180^\circ$ for $r \angle \theta$.) d) Which of the following two sinusoids are leading and by how many degrees? $v_1(t) = -2.5 \cos(\omega t + 70^\circ)$ V $v_2(t) = 3 \sin(\omega t - 70^\circ)$ V e) Find the sinusoid $i(t)$ of the current phasor $\bar{I}$, if $\bar{I} = j3e^{j15^\circ}$ A and $\omega = 50$ rad/s. Give the answer in cosine form with a positive amplitude and a phase angle between $\pm 180^\circ$.
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Vraag 2.5 [2]Bereken en gee die finale antwoord in **polêre formaat**.**Question 2.5 [2]**Solve and give final answer in **polar form**:

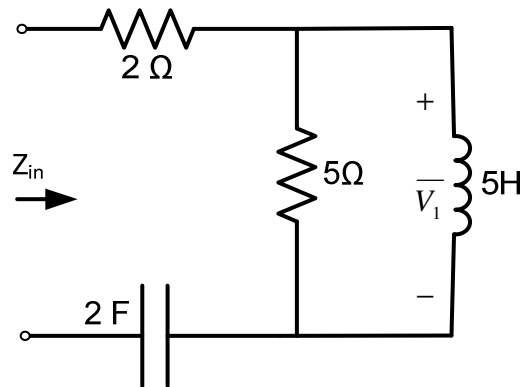
$$Z_1 = \frac{(2 + j10) + 2\angle(-180^\circ)}{8 + 7\angle 141.8^\circ}$$

Vraag 2.6 [2]

Bereken die impedansie Z_{in} indien $\omega=1\text{ rad/s}$. Skryf die antwoord in **reghoekige formaat**.

Question 2.6 [2]

Calculate the impedance Z_{in} if $\omega=1\text{ rad/s}$. Write the answer in **rectangular form**.

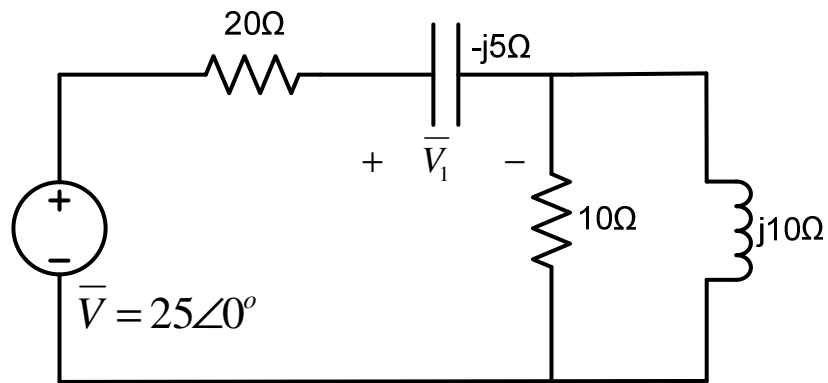


Vraag 2.7 [2]

Bereken die fasorspanning $\bar{V}_1$ oor die kapasitor en skryf die antwoord in **polêre formaat**.

Question 2.7 [2]

Determine the phasorvoltage $\bar{V}_1$ across the capacitor and write the answer in **polar form**.

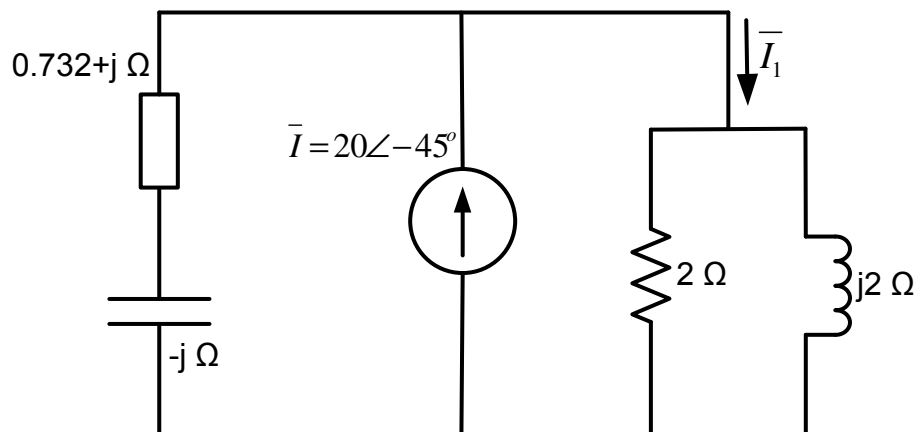


Vraag 2.8 [2]

Bepaal die fasorstroom $\bar{I}_1$ en gee die antwoord in **polêre formaat**.

Question 2.8 [2]

Find the phasor current $\bar{I}_1$ and give the answer in **polar form**.



EBN111 SEMESTER TOETS/TEST 2 (14 MAY 2014)**ANTWOORDBLAD / ANSWER SHEET**

Van / Surname:		Studentenommer: Student number:							
Voorletters / Initials:		Taal / Language	Afrikaans		English				

Q1		Q2		Tot:	/45
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VRAAG 1 – QUESTION 1 (25)

1.1	$V_{X(3A)} =$		
1.2	$I_Y =$		
1.3	$V_{TH} =$	$R_{TH} =$	
1.4	$I_N =$	$R_N =$	
1.5	$a) R_L =$	$b) P_{max} =$	
1.6	$a) V_X =$	$b) V_X =$	$c) V_X =$
1.7	$V_Y =$	$I_Y =$	
1.8	$V_Z =$	$I_Z =$	
1.9	$V_I =$	$V_2 =$	$I_I =$

VRAAG 2 – QUESTION 2 (20)

2.1	$C_{eq} =$		
2.2	$a) v_C =$	$b) w_L =$	
2.3	$a) i(t) =$	$b) w =$	
2.4	$a) f =$	$b) \phi =$	$c) \bar{V} =$
	$d) \text{_____ leads _____ by _____}$		$e) i(t) =$
2.5	$Z_I =$ (Polar format)		
2.6	$Z_{in} =$ (Rectangular format)		
2.7	$\bar{V}_1 =$ (Polar format)		
2.8	$\bar{I}_1 =$ (Polar format)		